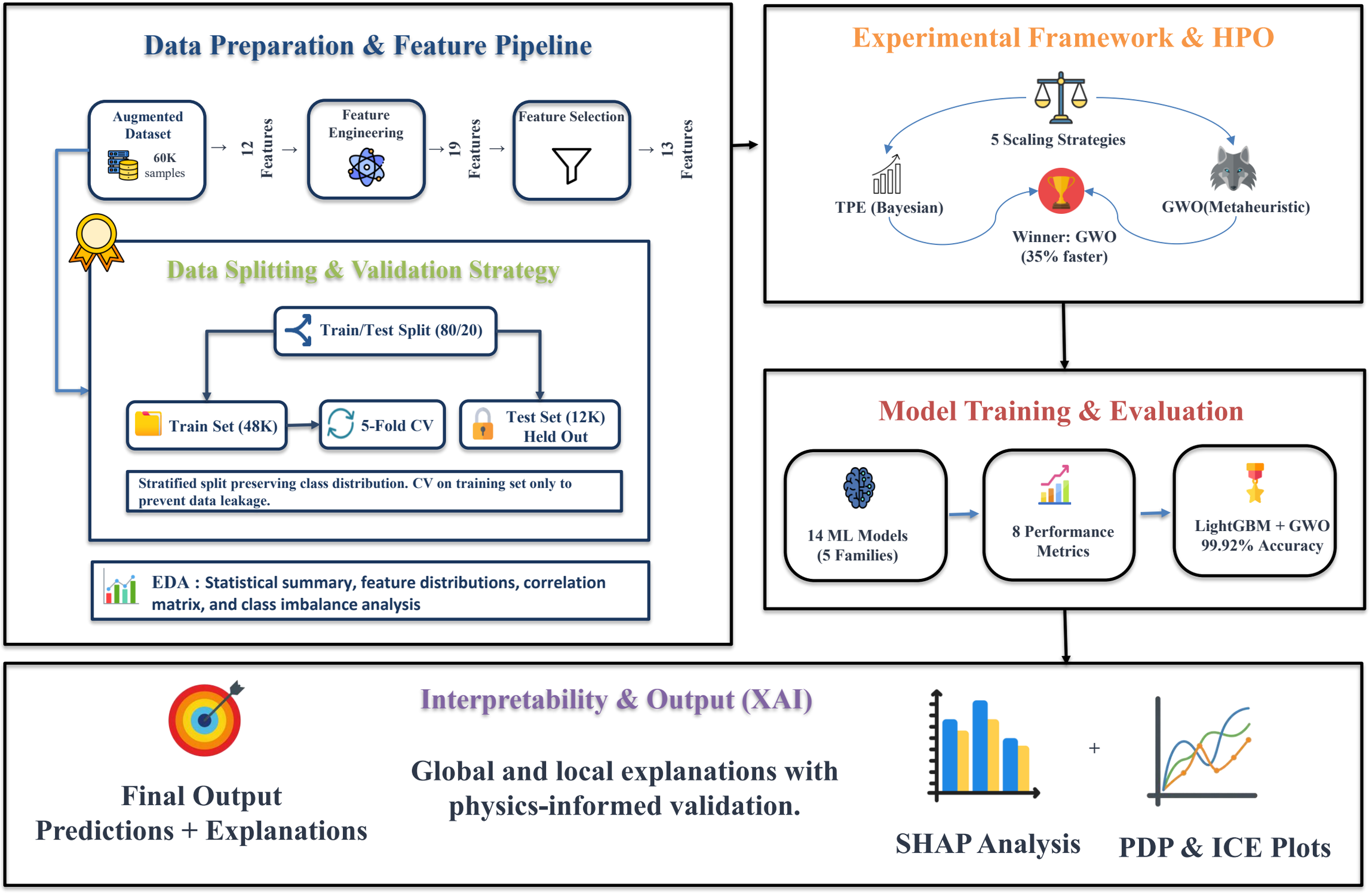
Data Preparation & Feature Pipeline
Augmented Dataset
60K samples
12 Features
Feature Engineering
19 Features
Feature Selection
13 Features
Data Splitting & Validation Strategy
Train/Test Split (80/20)
Train Set (48K)
5-Fold CV
Test Set (12K) Held Out
Stratified split preserving class distribution. CV on training set only to prevent data leakage.
EDA : Statistical summary, feature distributions, correlation matrix, and class imbalance analysis
Experimental Framework & HPO
5 Scaling Strategies
TPE (Bayesian)
GWO(Metaheuristic)
Winner: GWO (35% faster)
Model Training & Evaluation
14 ML Models (5 Families)
8 Performance Metrics
LightGBM + GWO 99.92% Accuracy
Final Output Predictions + Explanations
Interpretability & Output (XAI)
Global and local explanations with physics-informed validation.
SHAP Analysis
+
PDP & ICE Plots